CREDIT RISK MODELLING

Credit Risk Classification Using Logistic Regression: Predictive Modelling and Statistical Inference

A practical analysis of borrower characteristics, classification performance and model interpretation

DATASET South German Credit (UCI Machine Learning Repository)

OBSERVATIONS 1,000 applicants

METHOD Logistic regression with categorical encoding

EVALUATION Stratified train / validation / held-out test workflow

PROJECT REPORT • RESULTS AND INTERPRETATION • NO CODE

Executive Summary

This project develops a baseline logistic regression model to classify applicants according to the credit-risk label recorded in the South German Credit dataset. The analysis combines exploratory data analysis, held-out predictive evaluation, threshold selection and statistical interpretation of model coefficients.

Headline result

At the selected probability threshold of 0.34, the final model achieved 67.0% accuracy, 51.7% recall for the bad-risk class and a ROC-AUC of 0.698 on the held-out test set. These results indicate moderate ranking ability and a meaningful precision–recall trade-off, rather than a production-ready credit decision system.

Project element	Result
Dataset size	1,000 observations; 21 original columns
Outcome distribution	700 good-risk labels (70%); 300 bad-risk labels (30%)
Predictors used	Loan duration, credit amount, age, credit history, savings status and employment duration
Baseline model (threshold 0.50)	73.0% accuracy; 28.3% bad-risk recall; ROC-AUC 0.698
Final threshold	0.34 selected using validation data under a minimum 50% recall rule
Final held-out test performance	67.0% accuracy; 45.6% precision; 51.7% recall; 48.4% F1; ROC-AUC 0.698

The most consistent statistical association in the fitted inference model was loan duration: each additional month was associated with approximately 3.7% higher odds of the recorded bad-risk classification, conditional on the other included predictors ($p < 0.001$). Some categories of credit history, savings status and employment duration were also statistically significant relative to their reference categories. These are associations, not causal effects.

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1. Research Objective

The objective is to build and evaluate an interpretable baseline classifier that estimates whether an applicant belongs to the dataset's recorded bad-credit-risk category. The project asks three practical questions:

- How are applicants and the recorded risk labels distributed in the sample?
- How well does a logistic regression model distinguish the two recorded risk classes on data held out from model fitting?
- Which included borrower characteristics are associated with the outcome, and how does changing the probability threshold affect missed bad-risk cases and false alarms?

Scope clarification

The target is the dataset's recorded good/bad credit-risk classification. It is not a verified observation of whether a borrower subsequently defaulted on a loan.

2. Dataset and Variables

The South German Credit dataset contains 1,000 observations and 21 original variables. The data were loaded from the dataset's ASCII file and checked for shape, column names, target coding and missing values. The initial checks found 1,000 rows, 21 columns and zero missing values in the imported table.

Data check	Finding
Rows and columns	1,000 observations × 21 original variables
Missing values	0 in the imported dataset
Original outcome coding	kredit = 0: bad; kredit = 1: good
Project target recoding	default_risk = 1 for bad risk; 0 for good risk
Good-risk observations	700 (70%)
Bad-risk observations	300 (30%)

Selected predictors

Project variable	Meaning	Model treatment
Loan duration	Duration of credit in months	Numeric; standardised in predictive pipeline
Credit amount	Amount of credit requested	Numeric; standardised in predictive pipeline
Age	Applicant age in years	Numeric; standardised in predictive pipeline
Credit history	Recorded credit-history category	Categorical; one-hot encoded
Savings status	Recorded savings category	Categorical; one-hot encoded
Employment duration	Recorded employment-duration category	Categorical; one-hot encoded

Categorical codes were treated as categories rather than continuous quantities. This matters because the codes identify groups; the numerical distance between codes should not be assumed to represent a meaningful difference in risk.

3. Exploratory Data Analysis

3.1 Descriptive statistics

Statistic	Loan duration (months)	Credit amount	Age (years)
Mean	20.90	3,271.25	35.54
Standard deviation	12.06	2,822.75	11.35
Minimum	4	250	19
25th percentile	12	1,365.50	27
Median	18	2,319.50	33
75th percentile	24	3,972.25	42
Maximum	72	18,424	75

For both loan duration and credit amount, the mean exceeds the median, suggesting right-skewed distributions. The maximum loan duration is 72 months, compared with a median of 18 months. Credit amounts also vary widely, with a maximum of 18,424 against a median of 2,319.50. These differences motivate looking at group distributions rather than relying only on averages.

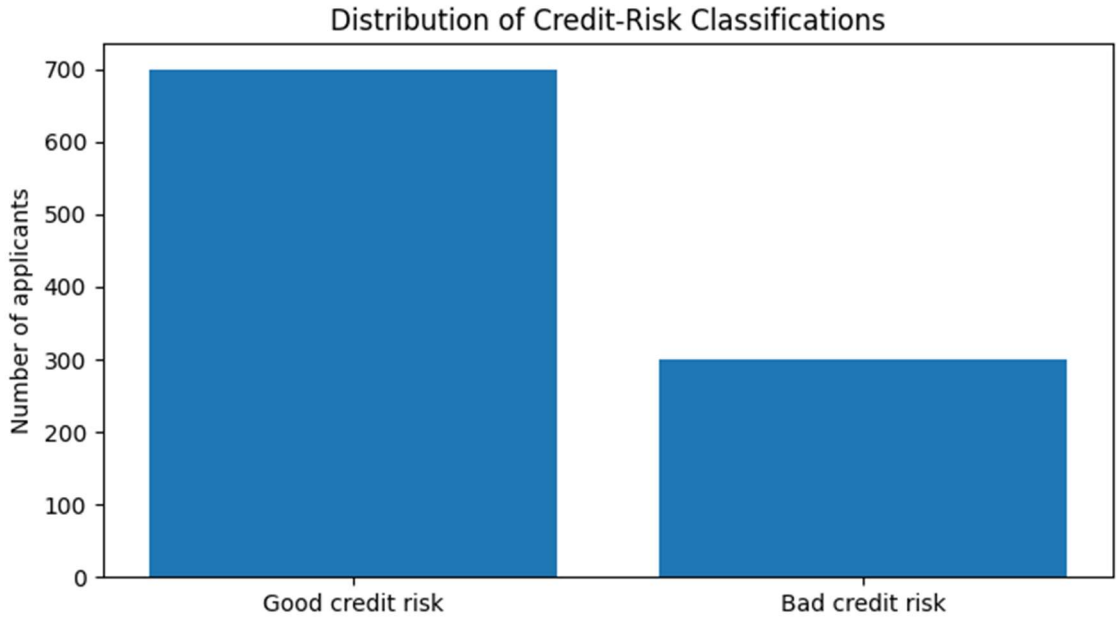


Figure 1. Recorded credit-risk class distribution: 700 good-risk and 300 bad-risk applicants.

The sample contains more good-risk than bad-risk labels. The 70:30 class distribution is important when interpreting accuracy: a classifier that always predicted the majority class would achieve 70% accuracy, although it would fail to identify any bad-risk cases.

3.2 Loan duration and credit amount by risk group

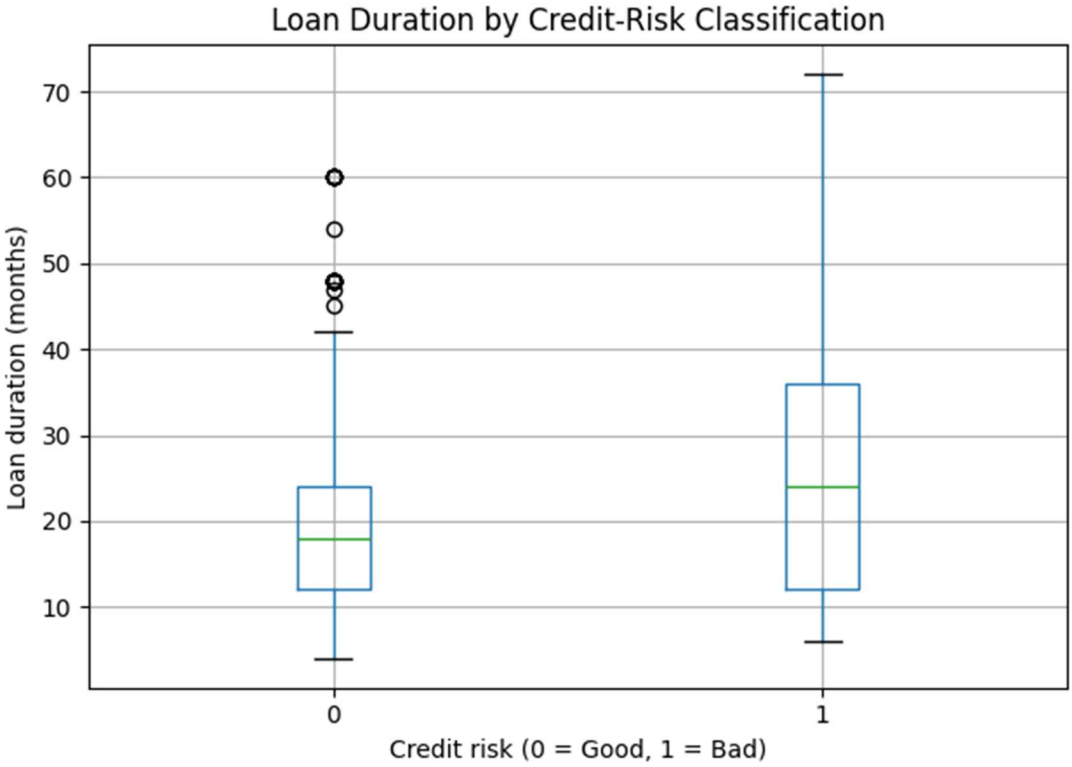


Figure 2. Loan-duration distributions by recorded risk class (0 = good; 1 = bad).

The boxplot suggests that the bad-risk group has a higher median loan duration and a wider spread. This is a descriptive pattern consistent with the positive loan-duration coefficient in the fitted logistic regression, but the plot alone does not establish a causal relationship.

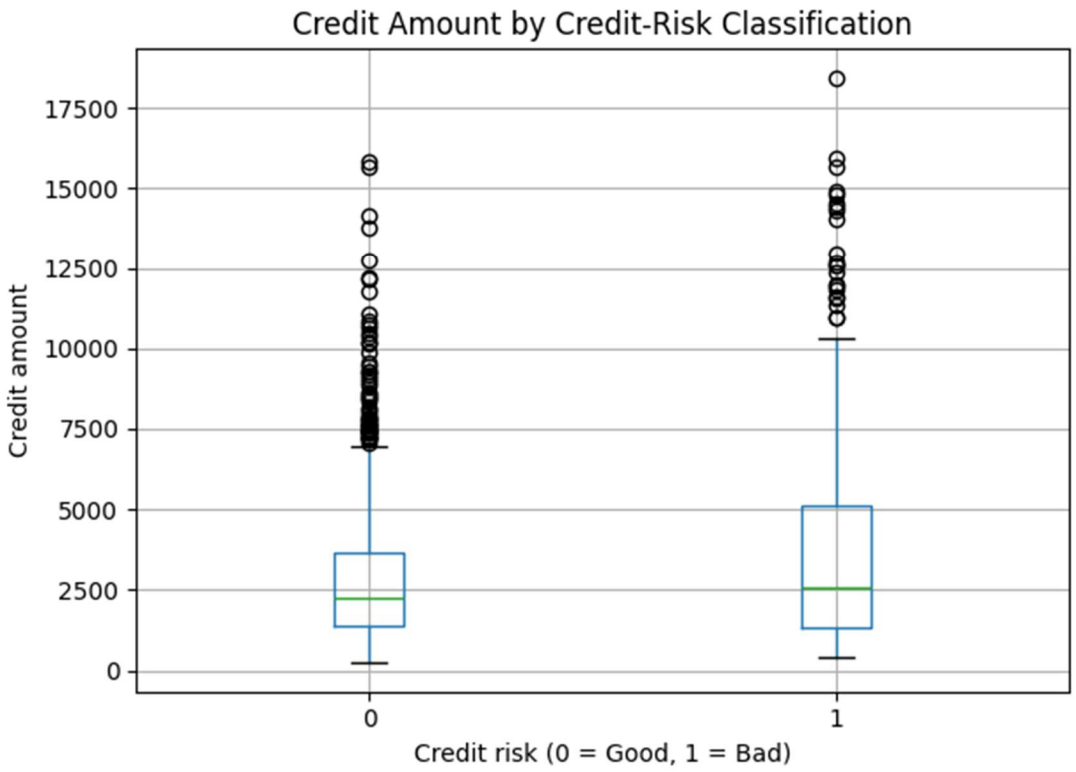


Figure 3. Credit-amount distributions by recorded risk class.

Credit amounts vary substantially in both groups, with many high-value observations. The distributions overlap, so credit amount alone is unlikely to separate the classes perfectly. In the fitted inference model, the credit-amount coefficient was not statistically significant at the 5% level.

3.3 Credit history and observed bad-risk rates

Credit-history code	Applicants	Observed bad-risk rate
0	40	62.50%
1	49	57.14%
2	530	31.89%
3	88	31.82%
4	293	17.06%

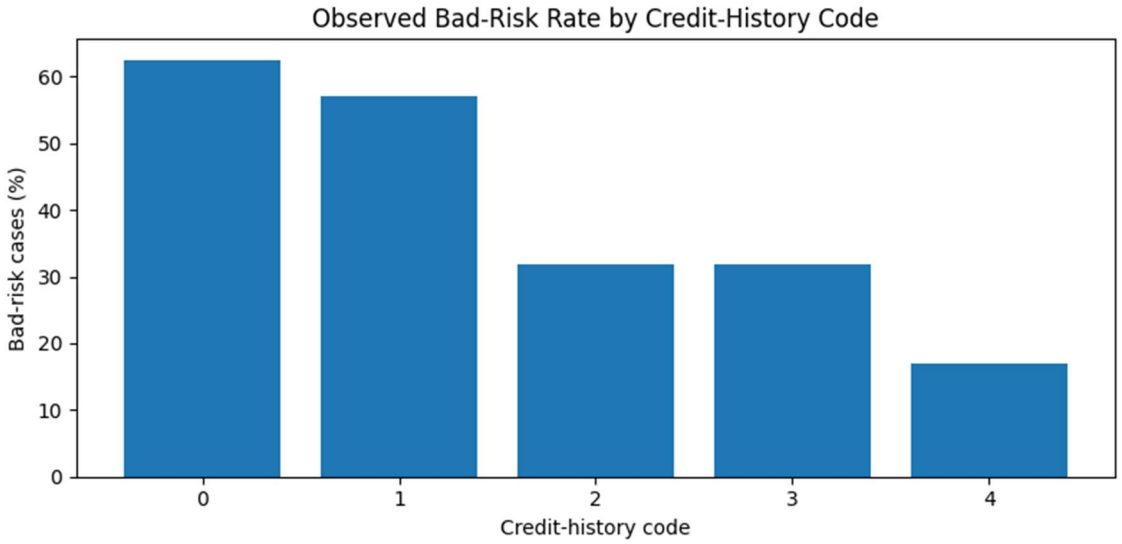


Figure 4. Observed bad-risk share by credit-history code.

Observed bad-risk proportions differ across credit-history categories in this sample. Code 0 has the highest observed rate (62.5%), while code 4 has the lowest (17.1%). The codebook defines these categories, so they should be interpreted as group labels rather than an automatically ordered numeric scale. Category sample sizes also differ, particularly for codes 0 and 1.

4. Modelling Approach

A logistic regression classifier was used because the target is binary. The original target was recoded so that 1 represents bad risk and 0 represents good risk. The selected predictors were three numerical variables and three categorical variables.

Stage	Procedure	Purpose
Train/test split	80% training and 20% test; stratified; random_state = 42	Preserve class proportions and reserve unseen test observations
Preprocessing	Standardise numerical variables; one-hot encode categorical variables	Place numeric predictors on comparable scales and represent categories appropriately
Model fitting	Logistic regression fitted on training data	Estimate class probabilities
Baseline classification	Threshold = 0.50	Establish a standard reference result
Threshold selection	Validation split from training data; tested thresholds; require recall ≥ 50%, then maximise precision	Choose a threshold without using the held-out test labels
Final evaluation	Refit on the full training set and evaluate on the held-out test set	Estimate performance on observations not used to fit the final model

The final threshold procedure split the original training set into a fitting subset and a validation subset. The selected threshold was 0.34, with validation precision of 52.9% and recall of 60.0%. The model was then refitted on the full training sample and evaluated on the held-out test set.

5. Baseline Model Performance

At the default threshold of 0.50, the baseline model achieved 73.0% accuracy and ROC-AUC of 0.698. However, its recall for the bad-risk class was only 28.3%, indicating that most bad-risk observations in the test sample were not identified at this threshold.

Metric	Baseline result	Meaning

Accuracy	73.0%	Share of all test observations classified correctly
Precision (bad risk)	60.7%	Among applicants flagged bad risk, share actually labelled bad risk
Recall (bad risk)	28.3%	Share of all actual bad-risk applicants identified
F1-score (bad risk)	38.6%	Harmonic mean of bad-risk precision and recall
ROC-AUC	0.698	Ranking discrimination across classification thresholds



Figure 5. Baseline confusion matrix at threshold 0.50 (test set, n = 200).

The baseline confusion matrix contains 129 correctly classified good-risk cases and 17 correctly classified bad-risk cases. It also contains 11 good-risk cases incorrectly flagged as bad risk and 43 bad-risk cases incorrectly classified as good risk. The latter are missed bad-risk classifications.

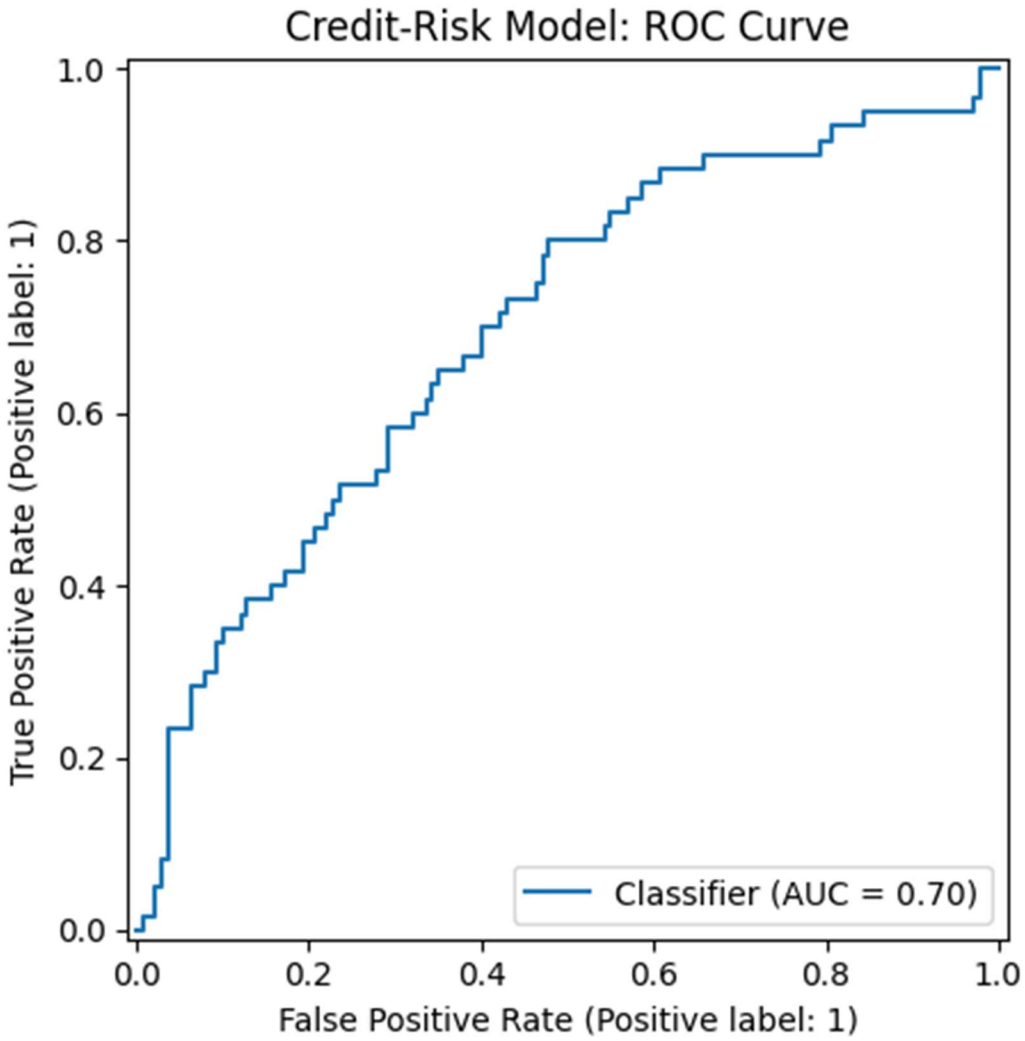


Figure 6. Baseline ROC curve; area under the curve (AUC) = 0.698.

A ROC-AUC of 0.698 indicates moderate discrimination: the model ranks a randomly selected bad-risk observation above a randomly selected good-risk observation more often than chance, but its ranking performance is far from perfect. Accuracy should be interpreted alongside recall and precision because the classes are imbalanced.

6. Coefficient Interpretation and Statistical Inference

A separate logistic regression was estimated on the 800-observation training sample using statsmodels, with categorical predictors explicitly represented as categories. This model provides coefficient estimates, standard errors, p-values and confidence intervals. Its numerical coefficients are not identical in scale to those from the standardised scikit-learn pipeline; the inference table below is interpreted using the original numerical units.

The categorical reference groups were credit-history code 0, savings-status code 1 and employment-duration code 1. Odds ratios for other categories compare each category with its respective reference group, holding the other included predictors constant.

Predictor / category	Odds ratio	95% CI for OR	p-value	Interpretation at 5%
Credit history: 1	1.695	0.623–4.614	0.3017	Not significant
Credit history: 2	0.500	0.232–1.081	0.0779	Not significant
Credit history: 3	0.355	0.143–0.886	0.0265	Lower odds vs. reference

Credit history: 4	0.191	0.083–0.441	0.0001	Lower odds vs. reference
Savings status: 2	0.703	0.416–1.187	0.1874	Not significant
Savings status: 3	0.285	0.124–0.654	0.0031	Lower odds vs. reference
Savings status: 4	0.248	0.082–0.750	0.0136	Lower odds vs. reference
Savings status: 5	0.273	0.162–0.458	<0.0001	Lower odds vs. reference
Employment duration: 2	0.835	0.398–1.752	0.6331	Not significant
Employment duration: 3	0.572	0.287–1.141	0.1130	Not significant
Employment duration: 4	0.366	0.171–0.782	0.0095	Lower odds vs. reference
Employment duration: 5	0.556	0.274–1.127	0.1033	Not significant
Loan duration (per month)	1.037	1.020–1.055	<0.0001	Higher odds per additional month
Credit amount (per unit)	1.000	approx. 1.000–1.0001	0.2781	Not significant
Age (per year)	0.993	0.976–1.010	0.4074	Not significant

Loan duration was positively associated with bad-risk classification: its odds ratio of 1.037 indicates approximately 3.7% higher odds per additional month, conditional on the other included predictors. The 95% confidence interval excludes 1 and the p-value is below 0.001. Some credit-history, savings-status and employment-duration categories also differ significantly from their reference categories.

Credit amount and age were not statistically significant at the 5% level in this specification. For credit amount, the odds ratio per 1,000 units was 1.041, with a 95% confidence interval of 0.968–1.120 and p = 0.2781. Because the interval includes 1, the model does not provide strong evidence of an association at the 5% level.

Model fit statistics

The training-sample inference model used 800 observations, with McFadden pseudo-R^2 = 0.1427 and log-likelihood = −418.967. Pseudo-R^2 is not the same as ordinary least-squares R^2 and should not be described as the percentage of variance explained.

7. Threshold Selection and Final Evaluation

Logistic regression returns probabilities; a classification threshold converts those probabilities into class labels. Lower thresholds tend to identify more bad-risk cases (higher recall), but they also tend to flag more good-risk cases incorrectly and reduce precision. The threshold should reflect the intended use and the relative costs of the two types of error.

7.1 Exploratory threshold comparison

Threshold	Accuracy	Bad-risk precision	Bad-risk recall	Bad-risk F1	Flagged bad risk
0.30	64.5%	43.8%	65.0%	52.3%	89
0.35	69.0%	48.4%	51.7%	50.0%	64
0.40	69.0%	48.1%	41.7%	44.6%	52
0.45	72.5%	56.8%	35.0%	43.3%	37
0.50	73.0%	60.7%	28.3%	38.6%	28

This comparison is exploratory because it used predictions on the test set. It is useful for illustrating the trade-off, but it should not be used to claim a final optimised threshold or unbiased performance estimate.

7.2 Final threshold selected using validation data

Selected threshold: 0.34

The validation procedure selected 0.34 as the highest-precision candidate among the tested thresholds that achieved at least 50% validation recall. Validation precision was 52.9% and validation recall was 60.0%.

Metric	Baseline threshold 0.50	Final threshold 0.34
Accuracy	73.0%	67.0%
Bad-risk precision	60.7%	45.6%
Bad-risk recall	28.3%	51.7%
Bad-risk F1-score	38.6%	48.4%
ROC-AUC	0.698	0.698



Figure 7. Final held-out test confusion matrix at the validation-selected threshold of 0.34.

At threshold 0.34, the model correctly classified 103 good-risk cases and 31 bad-risk cases. It incorrectly flagged 37 good-risk cases as bad risk and missed 29 bad-risk cases. Compared with the baseline, bad-risk recall increased from 28.3% to 51.7%, while accuracy fell from 73.0% to 67.0%. ROC-AUC remained 0.698 because changing the threshold does not change the underlying probability ranking.

This is a deliberate trade-off rather than an unambiguous improvement: the model finds more bad-risk cases, but a larger share of its flags are false positives. A real lending decision would require a documented cost framework and additional validation before choosing a threshold.

8. Limitations and Conclusion

Limitations

- The dataset contains only 1,000 observations and may not represent a current lending population or the conditions of a particular financial institution.

- The target is the recorded good/bad credit-risk label, not verified subsequent defaults or losses. The model should therefore be described as a credit-risk classification model.
- The six predictors are a deliberately small subset of the available variables. Other predictors or alternative specifications could change estimates and predictive performance.
- The final ROC-AUC of 0.698 indicates moderate discrimination, and the final precision of 45.6% means that fewer than half of the applicants flagged as bad risk in the test sample were labelled bad risk.
- Coefficient p-values and confidence intervals depend on model assumptions and specification. Statistical association does not establish causality.
- This is an educational baseline, not a production credit-scoring system. It has not been assessed for fairness, stability over time, calibration, regulatory compliance or real-world financial costs.

Conclusion

This project demonstrates an end-to-end introductory credit-risk classification workflow: data validation, exploratory analysis, categorical preprocessing, logistic regression, held-out evaluation, coefficient inference and validation-based threshold selection. The model achieved a ROC-AUC of 0.698 on the held-out test sample. Selecting a threshold of 0.34 increased bad-risk recall to 51.7%, at the cost of lower precision and accuracy. Loan duration showed a positive, statistically significant association with the recorded bad-risk label, while selected categories of credit history, savings status and employment duration also showed statistically significant differences relative to their reference groups.

The principal lesson is that credit-risk modelling involves more than maximising accuracy. Model performance must be assessed using metrics that reflect the outcome of interest, and threshold decisions should be informed by the consequences of missed risky applicants and false alarms. Further work could compare a broader predictor set, assess probability calibration, perform cross-validation and evaluate fairness and stability before any practical deployment.

Appendix: Key Results at a Glance

Item	Final recorded result
Dataset	South German Credit; 1,000 observations
Outcome labels	700 good risk; 300 bad risk
Predictors	Loan duration, credit amount, age, credit history, savings status, employment duration
Inference sample	800 training observations
McFadden pseudo-R^2	0.1427
Significant individual terms at 5%	Credit-history categories 3 and 4; savings categories 3, 4 and 5; employment-duration category 4; loan duration
Validation-selected threshold	0.34
Final test accuracy / precision / recall / F1	67.0% / 45.6% / 51.7% / 48.4%
Final test ROC-AUC	0.698